

Developer Mock Test Results

Student: Shiva kasula

Name:	Shiva kasula	Student ID:	112
Date:	2026-03-05 11:53:44	Performance:	Needs Improvement
Overall Score:	9.8/25 (39.2%)	Test ID:	7c7e14d1-5081-4274...
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	5/10	50.0%	Pass
MCQ/Theory	0/10	0.0%	Needs Work
Coding	4.8/5	96.0%	Pass

Detailed Question Review

APTITUDE SECTION (5/10 - 50.0%)

Q1. What is the average of 2, 4, 6, 8, 10?

Your Answer: 6

Excellent! Right answer.

Q2. A product is on sale for 25% off its original price of \$100. How much will you pay for the product?

Your Answer: \$80

Correct Answer: \$75

To find the sale price, calculate 25% of the original price of \$100 and subtract it from \$100. The key calculation step is: $\$100 \times 0.25 = \25 , then $\$100 - \$25 = \$75$. The user's likely mistake was incorrect calculation of the discount, as they answered \$80 instead of the correct result of \$75.

Q3. A mixture of juice and water is in the ratio 4:5. If 5 cups of juice are added to the mixture, what is the new ratio of juice to water?

Your Answer: 5:4

Correct Answer: 9:5

The correct answer is 9:5 because when 5 cups of juice are added to the original mixture with a ratio of 4:5, the amount of juice increases significantly. To find the new ratio, let's assume the original mixture had 4x cups of juice and 5x cups of water, and after adding 5 cups of juice, the new amount of juice is 4x + 5, and the amount of water remains 5x. The key calculation step is to find the least common multiple or a common basis for comparison, but since the question asks for a ratio, we can directly compare the increase in juice to the original amounts. The user's likely mistake is not accounting for the original ratio and the increase in juice properly, as they chose 5:4, which would be correct if the original ratio was reversed or if the increase in juice was not properly calculated.

■ Q4. If 18 workers can build a house in 12 days, how many workers will be needed to build the same house in 10 days?

Your Answer: 24 workers

Correct Answer: 22 workers

■ To find the number of workers needed to build the house in 10 days, we can use the formula: (number of workers) * (number of days) = constant. Given that 18 workers can build the house in 12 days, we have $18 * 12 = 216$ worker-days. To build the same house in 10 days, we need $216 \text{ worker-days} / 10 \text{ days} = 21.6$ workers. The user's answer of 24 workers is likely an overestimation. The correct calculation yields approximately 21.6 workers, which is closest to option D) 21 workers, but since the question likely seeks a whole number and given the provided options, the closest higher whole number is 22 workers, as the work cannot be done with a fraction of a worker and more workers are needed to complete the task in less time.

■ Q5. If a number is increased by 25% and then decreased by 25%, what is the overall percentage change?

Your Answer: No answer (Skipped)

Correct Answer: -6.25%

■ To find the overall percentage change, let's consider a number, say 100. When increased by 25%, it becomes $100 + (100 * 0.25) = 125$. Then, when decreased by 25%, it becomes $125 - (125 * 0.25) = 93.75$. The key calculation step is: $(100 * 0.25) = 25$ for the increase, and $(125 * 0.25) = 31.25$ for the decrease. The correct answer is -6.25% because the final value (93.75) is less than the original value (100), indicating a decrease. The user likely got it wrong if they mistakenly thought the increase and decrease would cancel each other out, not accounting for the different bases (100 and 125) used for the percentage calculations.

■ Q6. A shopkeeper buys an item at Rs.150 and sells it at a 12% profit. What is the selling price?

Your Answer: Rs.168

■ Excellent! Right answer.

■ Q7. A mixture is made of 2 parts acid and 3 parts water. What is the ratio of acid to the total mixture?

Your Answer: 2/5

■ Correct! Well done.

■ Q8. If 8 workers can build a house in 12 days, how many days will it take for 6 workers to build the same house?

Your Answer: 16 days

■ Correct! Well done.

■ Q9. A store buys a product for \$40 and sells it for \$35. What is the loss percentage?

Your Answer: 10%

Correct Answer: 12.5%

■ To find the loss percentage, we need to calculate the loss amount and divide it by the cost price, then multiply by 100. The key calculation step is: $\text{Loss percentage} = ((\text{Cost price} - \text{Selling price}) / \text{Cost price}) * 100 = ((40 - 35) / 40) * 100 = (5 / 40) * 100 = 12.5\%$. The user's answer of 10% is likely incorrect because they may have made a calculation error or misunderstood the formula for loss percentage.

■ Q10. A store buys a product for \$30 and sells it for \$25. What is the loss percentage?

Your Answer: 16.67%

■ Correct! Well done.

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What is profiling in Java?

Your Answer:

The process of analyzing the functionality of a program

Correct Answer:

The process of analyzing the performance of a program

■ The correct answer is A) The process of analyzing the performance of a program, because profiling in Java refers to the process of analyzing and measuring the performance of a Java program, identifying areas that consume the most resources, such as CPU time and memory. The user's answer, B) The process of analyzing the functionality of a program, is incorrect because it describes a different aspect of program analysis, focusing on how the program works rather than how efficiently it works. The user's mistake likely stems from confusing profiling with other types of program analysis, such as debugging or testing.

■ Q2. What are objects in Java?

Your Answer: Methods

Correct Answer: Instances of classes

■ The correct answer, "Instances of classes", is right because in Java, an object is an instance of a class, which has its own set of attributes (data) and methods (functions). This means that an object represents a real-world entity or concept, and it has its own state and behavior. The user's answer, "Methods", is likely incorrect because methods are functions that belong to a class or object, but they are not the objects themselves.

■ Q3. What is concurrency in Java?

Your Answer: The ability of a program to execute multiple threads or processes sequentially

Correct Answer: The ability of a program to execute multiple threads or processes concurrently

■ The correct answer is A) The ability of a program to execute multiple threads or processes concurrently, because concurrency in Java refers to the ability of a program to execute multiple threads or processes simultaneously, improving responsiveness and throughput. The user's answer, C) The ability of a program to execute multiple threads or processes sequentially, is incorrect because it describes a sequential execution, which is the opposite of concurrency. The user's likely mistake is misunderstanding the concept of concurrency, which involves simultaneous execution, not sequential execution.

■ Q4. What is used to handle exceptions in Java?

Your Answer: If-else statement

Correct Answer: Try-catch block

■ The correct answer is the try-catch block because it is a specific Java construct designed to handle exceptions, allowing developers to enclose code that might throw an exception and provide a way to catch and handle it. The user's answer, "If-else statement", is incorrect because while if-else statements can be used for conditional logic, they are not specifically designed for exception handling. The user likely mistook the purpose of if-else statements, which are used for controlling program flow based on conditions, not for handling exceptions.

■ Q5. What is an example of a method in Java?

Your Answer: A method that takes a boolean and returns its opposite

Correct Answer: All of the above

■ The correct answer is "All of the above" because each option describes a valid method in Java. A method in Java is a block of code that can take parameters, perform operations, and return a result, which is exactly what options A, B, and C describe. The user's likely mistake was not considering that a method can be defined with various parameters and return types, such as numbers, strings, and booleans, making all options valid examples of methods in Java.

■ Q6. What is a benefit of using Java?

Your Answer: Fast execution

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java is indeed easy to learn due to its simple syntax, it has fast execution thanks to the Just-In-Time (JIT) compiler, and it is reliable and secure with its built-in memory management and robust security features. The user's answer, "Fast execution", is partially correct but incomplete, as it only highlights one of the benefits of using Java. The user's likely mistake is not considering the other advantages of Java, such as its ease of learning and reliability, which are also significant benefits.

■ Q7. What is the next step after creating a new Java file?

Your Answer: Open a code editor or IDE

Correct Answer: Write your code

■ The correct answer is "Write your code" because after creating a new Java file, the next logical step is to start writing the code that will be executed by the Java program. The user's answer, "Open a code editor or IDE", is likely incorrect because

this step would have already been completed in order to create the new Java file. The user probably confused the steps involved in starting a new Java project, as opening a code editor or IDE is a prerequisite to creating a new Java file, not a subsequent step.

■ Q8. What is a characteristic of Java programs?

Your Answer:

They are only used for web development

Correct Answer:

They are easy to maintain due to their structured and class-based design

■ The correct answer, "They are easy to maintain due to their structured and class-based design", is right because Java's object-oriented design and structured syntax make it easier for developers to organize, modify, and scale their code. This design also enables better code reusability and readability. The user's answer, "They are only used for web development", is incorrect because Java is a versatile language used in various applications, including mobile apps, desktop applications, and enterprise software, not just web development.

■ Q9. What is polymorphism in Java?

Your Answer: The ability of an object to take on a single form

Correct Answer: The ability of an object to take on multiple forms

■ The correct answer is A) The ability of an object to take on multiple forms, because polymorphism in Java refers to the ability of an object to behave like multiple types, depending on the context in which it is used. This allows for more flexibility and generic code, as objects of different classes can be treated as if they were of the same class. The user's mistake likely stems from misunderstanding the concept of polymorphism, which is often confused with having a single, fixed form, rather than the ability to adapt to different forms.

■ Q10. What should variable and method names be in Java?

Your Answer: Meaningless

Correct Answer: Meaningful

■ The correct answer is "Meaningful" because in Java, variable and method names should be descriptive and convey their purpose to make the code readable and understandable. This is a fundamental principle of good coding practice, allowing other developers to easily comprehend the code's functionality. The user's answer, "Meanless", is likely a mistake because it goes against this principle, suggesting that names should be random or unclear, which would hinder code readability and maintainability.

■ CODING SECTION (4.8/5 - 96.0%)

■ Q1. Write a program that reads a string and prints the string with all characters converted to uppercase.

****Input Format:**** - Line 1: Input string (strin...

Your Answer:

```
import java.util.Scanner;

public class UpperCaseString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read input string
        String input = sc.nextLine();

        // Convert to uppercase
        String upperCase = input.toUpperCase();

        // Print result
        System.out.println(upperCase);

        sc.close();
    }
}
```

```
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on achieving an overall "Accepted" status, keep up the good work!

■ **Q2. Write a Java program that reads a queue of integers from stdin and finds the maximum element.**

****Input Format:** - Line 1: The number of elements in t...**

Your Answer:

```
import java.util.Scanner;
import java.util.Queue;
import java.util.LinkedList;

public class MaxInQueue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read the number of elements
        int n = sc.nextInt();
        sc.nextLine(); // Consume the leftover newline

        // Read the queue elements
        Queue<Integer> queue = new LinkedList<>();
        for (int i = 0; i < n; i++) {
            queue.add(sc.nextInt());
        }

        // Find the maximum element
        int max = Integer.MIN_VALUE;
        for (int num : queue) {
            if (num > max) {
                max = num;
            }
        }

        // Print the maximum
        System.out.println(max);

        sc.close();
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q3. Write a program that reads an integer n and prints the first n numbers in the Fibonacci sequence.**

****Input Format:** - Line 1: n (integer) **Output F...**

Your Answer:

```
import java.util.Scanner;

public class FibonacciSequence {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read the number of Fibonacci numbers to print
        int n = sc.nextInt();

        int a = 0, b = 1;

        for (int i = 0; i < n; i++) {
            System.out.print(a + " ");
        }
    }
}
```

```

        int next = a + b;
        a = b;
        b = next;
    }

    sc.close();
}
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing accurate and effective code that meets the requirements.

■ **Q4. Write a program that simulates a bank account, allowing users to deposit, withdraw, and check their balance. ***Input Format:*** - Line 1: Initial bala...**

Your Answer:

```

import java.util.Scanner;

public class BankAccountSimulation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read initial balance
        int balance = sc.nextInt();

        // Read number of transactions
        int n = sc.nextInt();
        sc.nextLine(); // Consume leftover newline

        // Process each transaction
        for (int i = 0; i < n; i++) {
            String transactionType = sc.next();
            int amount = sc.nextInt();

            if (transactionType.equalsIgnoreCase("deposit")) {
                balance += amount;
            } else if (transactionType.equalsIgnoreCase("withdraw")) {
                balance -= amount;
            }
        }

        // Print final balance
        System.out.println(balance);

        sc.close();
    }
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q5. Write a Java program that reads a queue of integers from stdin and finds the sum of its elements. ***Input Format:*** - Line 1: The number of elements ...**

Your Answer:

```

import java.util.Scanner;
import java.util.Queue;
import java.util.LinkedList;

public class SumOfQueue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

// Read number of elements
int n = sc.nextInt();
sc.nextLine(); // Consume newline

// Read queue elements
Queue<Integer> queue = new LinkedList<>();
for (int i = 0; i < n; i++) {
    queue.add(sc.nextInt());
}

// Calculate sum
int sum = 0;
for (int num : queue) {
    sum += num;
}

// Print the sum
System.out.println(sum);

sc.close();
}
}

```

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over
        String[] queueElements = scanner.nextLine().split(" ");
        int sum = 0;
        for (String element : queueElements) {
            sum += scanner.nextInt();
            scanner.nextLine(); // Consume newline left-over
        }
        System.out.println(sum);
        scanner.close();
    }
}

```

■ The student's code failed to produce the expected output because it did not calculate the sum of the queue elements. The example output shows that the expected sum is 15 (1+2+3+4+5), but the code only reads the elements into a queue without summing them. The actual output of 1 suggests that the code may be printing the first element instead of the sum.

■ Recommendations

Areas that need improvement:

- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.